



Department of Computer Science and Engineering (Data Science)

Name: Bhuvi Ghosh

SAP ID:60009210191

DIV: B

Lab Manual

Subject: Foundations of Data Analysis Laboratory (DJ19DSL303)

Semester: III

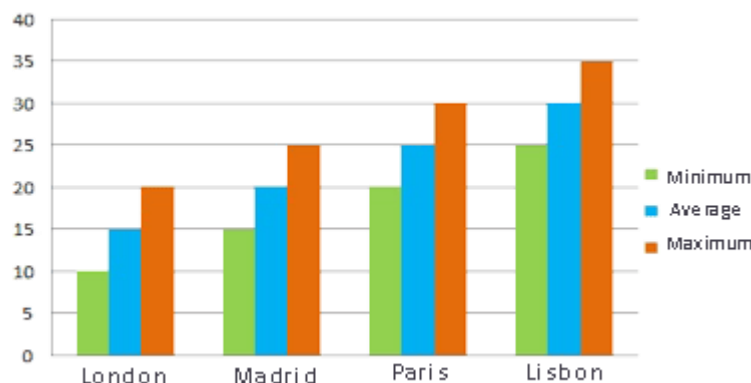
Experiment 1

(Data Visualization)

Aim: Create new measures on a given dataset and visualize them using a bar graph.

Theory:

A bar chart or bar graph is a chart or graph that presents categorical data with rectangular bars with heights or lengths proportional to the values that they represent. The bars can be plotted vertically or horizontally. A vertical bar chart is sometimes called a column chart.



A Grouped Bar Graph denoting the temperatures in different cities on a specific day

A bar graph shows comparisons among discrete categories. One axis of the chart shows the specific categories being compared, and the other axis represents a measured value. Some bar graphs present bars clustered in groups of more than one, showing the values of more than one measured variable.

Bar graphs/charts provide a visual presentation of categorical data. Categorical data is a grouping of data into discrete groups, such as months of the year, age group, shoe sizes, and animals. These categories are usually qualitative. In a column (vertical) bar chart, categories appear along the horizontal axis and the height of the bar corresponds to the value of each category.

Bar charts have a discrete domain of categories and are usually scaled so that all the data can fit on the chart. When there is no natural ordering of the categories being compared, bars on the chart may be arranged in any order. Bar charts arranged from highest to lowest incidence are called Pareto charts.



Department of Computer Science and Engineering (Data Science)

Grouped (clustered) and stacked:

In grouped (clustered) bar charts, for each categorical group there are two or more bars color-coded to represent a particular grouping. For example, a business owner with two stores might make a grouped bar chart with different colored bars to represent each store: the horizontal axis would show the months of the year and the vertical axis would show revenue.

Alternatively, a stacked bar chart stacks bars on top of each other so that the height of the resulting stack shows the combined result. Stacked bar charts are not suited to data sets having both positive and negative values.

Lab Assignments to complete in this session

Use the given dataset and perform the following tasks:

Dataset:

https://raw.githubusercontent.com/CSSEGISandData/COVID-19/web-data/data/cases_country.csv

Compatible Web Data Connector for easy importing:

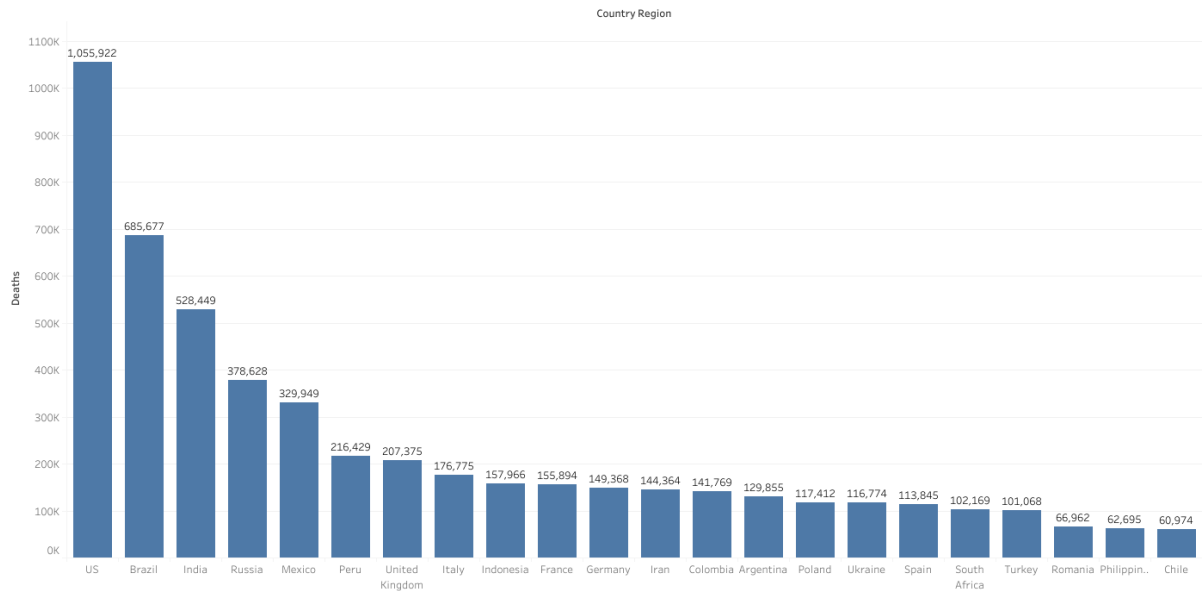
<https://basic-csv-wdc.herokuapp.com/>



Department of Computer Science and Engineering (Data Science)

- a) The United Nations (UN) wants to know what Countries are suffering more deaths due to COVID-19, in order to donate additional vaccines and prevent the spread of COVID, from the additional funds and vaccine budget of this month.

Sheet 1



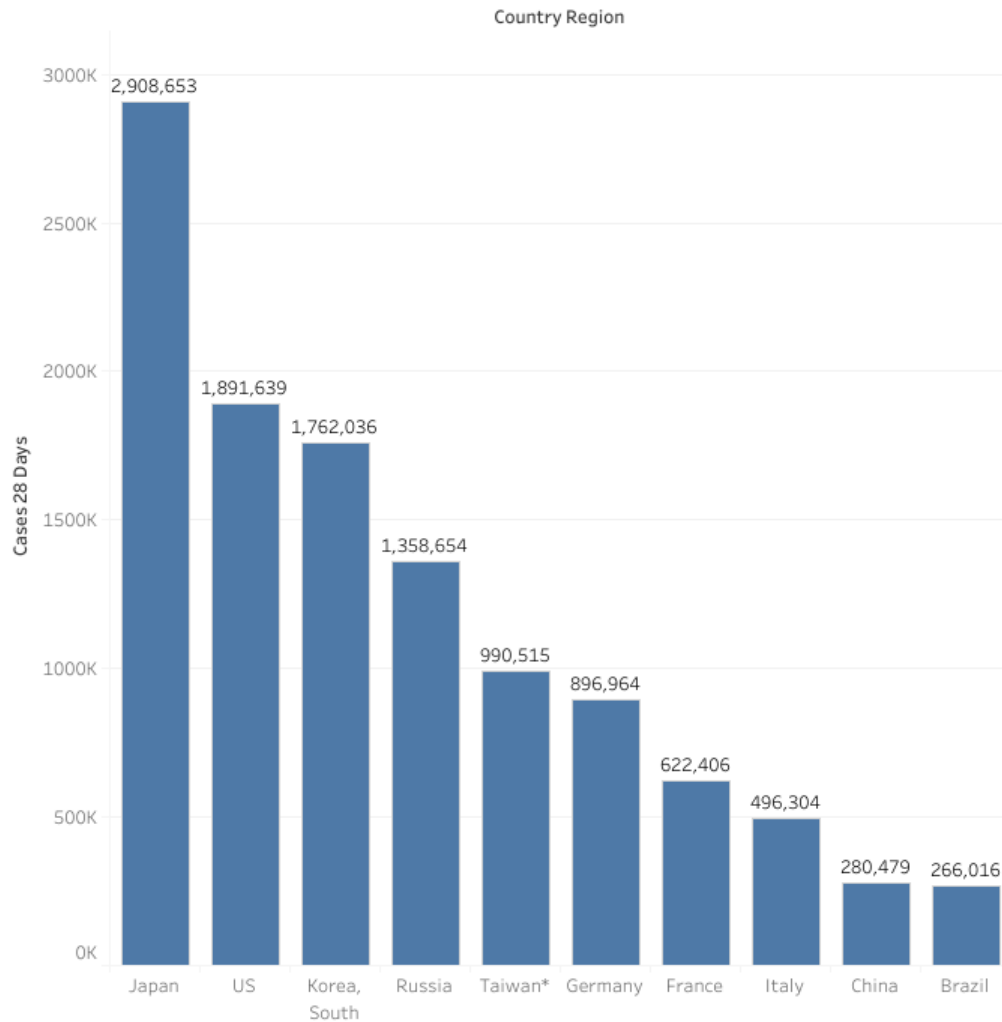
After analysing the above data using Tableau, it is found that USA followed by Brazil ,India ,Russia & Mexico are suffering more deaths due to Covid-19.



Department of Computer Science and Engineering (Data Science)

- b) Find out what are the top 10 Countries suffering severely due to COVID-19 in the past month eligible for additional vaccine donations.

Sheet 2



The top 10 Countries suffering severely due to COVID-19 in the last 28 days are:

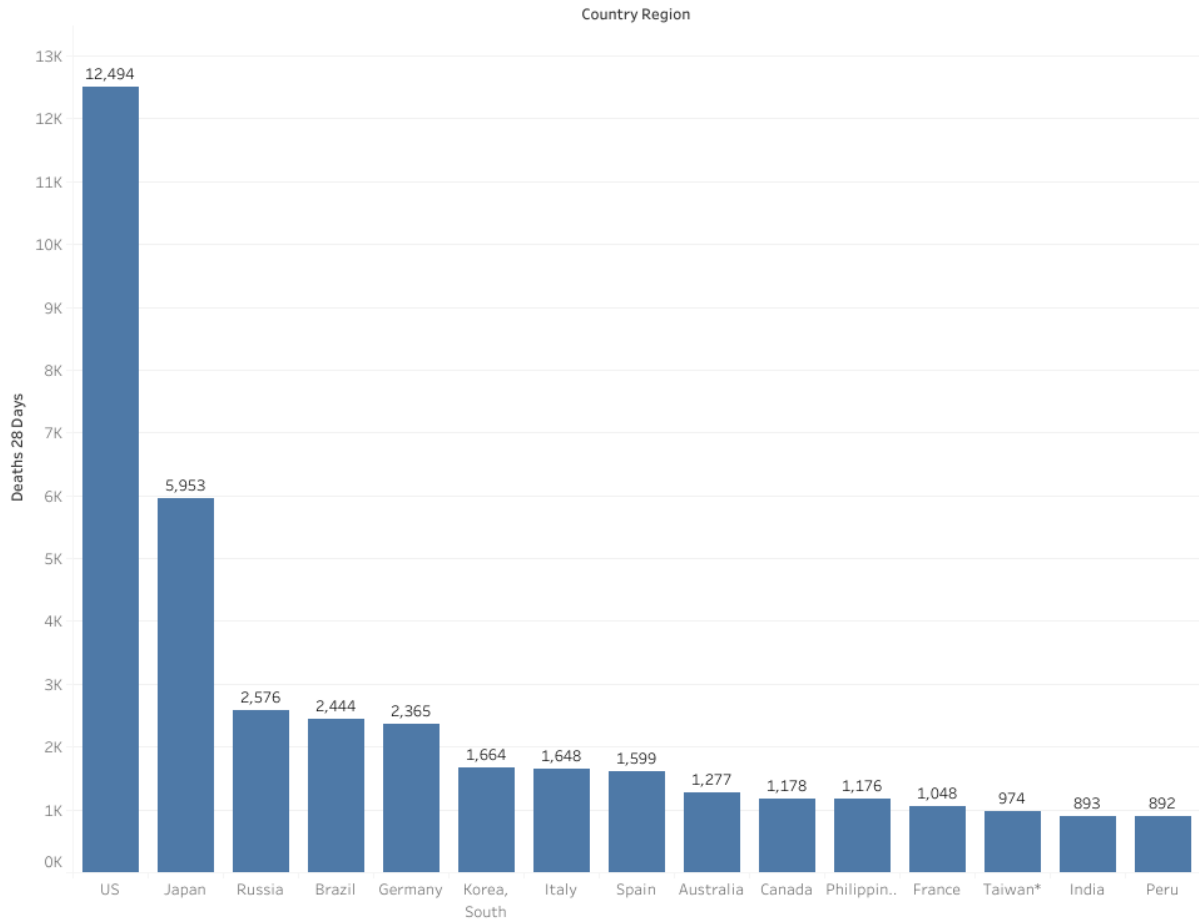
- 1) Japan*
- 2) US*
- 3) South Korea*
- 4) Russia*
- 5) Taiwan*
- 6) Germany*
- 7) France*
- 8) Italy*
- 9) China*
- 10) Brazil*



Department of Computer Science and Engineering (Data Science)

- c) Need for vaccines and funding are estimated from the number of deaths in the past month.

Sheet 3



USA has the largest need for vaccine & funding as it has suffered from the most number of deaths in the last 28 days.

Conclusion: The given dataset has been visualised using a bar graph and the necessary inferences are made out of it.